

# Distance Decay as a Primary Transferable Component: Zero-Shot Gravity Model Calibration from Trip-Length Distributions

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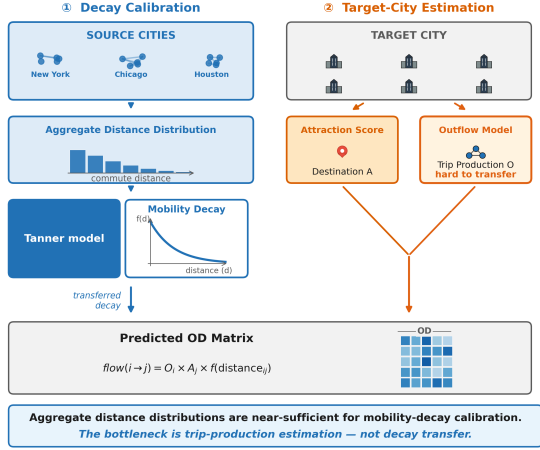
## Abstract

Predicting commuting flows in cities without access to individual travel trajectories is a central challenge in urban mobility modeling: most high-accuracy methods require local origin-destination (OD) surveys for calibration, limiting their applicability in data-scarce or privacy-regulated environments. This study investigates whether trip-length distributions (TLDs)—population-level histograms of traveled distances available from mobile operators and tech platforms—contain sufficient information to recover the city-specific distance-decay structure for OD-free gravity model calibration. We distinguish two transfer settings: an *OD-free calibration* setting that uses only target-city TLDs (but not individual OD records), and a *pure zero-shot fallback* that uses no target mobility data whatsoever. Evaluating across 50 US metropolitan areas (25 source cities for training and 25 held-out target cities for evaluation), we show that decay parameters recovered from TLD histograms alone correlate strongly with the power-law exponent  $\gamma$  fitted from complete OD matrices ( $r = \mathbf{0.909}$ ), and show weak-to-moderate correlation for the exponential rate  $\beta$  ( $r = \mathbf{0.427}$ ). Under synthetic 20-bin oracle outflow conditions, using recovered parameters yields a negligible downstream prediction gap. Through factorial ablation and game-theoretic Shapley value analysis, we show that distance decay dominates predictive capacity specifically within the investigated gravity framework, indicating it functions as the primary transferable structural component of spatial interaction within this structured model class (accounting for 85.8% of the total gain). Under oracle outflow conditions (where target outflow totals are controlled), the proposed model using target-city Meta MDM decay achieves downstream performance statistically equivalent to the locally-calibrated Traditional Tanner gravity baseline within a tight 0.01 CPC margin; however, under GBDT predicted outflows, performance is bounded by production prediction errors, yielding a minor performance drop that is not statistically equivalent. In the fully survey-free 25-city evaluation using independently sourced Meta MDM decay, the proposed model remains below DeepGravity but provides a highly interpretable and data-efficient OD-free alternative.

**Keywords:** Urban mobility, distance decay calibration, gravity models, aggregate mobility data, zero-shot transfer, open-data modeling

## 1 Introduction

Predicting origin-destination (OD) travel flows is essential for urban transport planning and infrastructure design (Barbosa et al. 2018; González



**Fig. 1** Overview of the core findings: Aggregate trip-length distributions contain sufficient information to recover city-specific distance decay  $f(d)$ , which acts as a primary transferable structural component for predicting zero-shot flows within the investigated gravity framework. Prediction performance is bounded by the model’s structural gravity assumptions.

et al. 2008; Song et al. 2010). However, traditional travel surveys and proprietary cellular records are highly restricted due to cost, commercial licensing, and privacy concerns. Consequently, a central challenge in urban mobility modeling is the zero-shot cross-city transfer problem: predicting commuting flows in a target city where no local OD surveys are available.

While recent deep learning approaches achieve high accuracy, they require abundant individual-level OD records for training, limiting their generalizability and raising privacy concerns. In contrast, classical gravity models (Wilson 1971) offer physical interpretability and structural simplicity but require local calibration of their distance-decay functions. This raises a fundamental scientific question: **Can aggregate mobility distributions, which do not reveal individual OD pairs, serve as a reliable calibration signal for recovering city-specific distance-decay functions?**

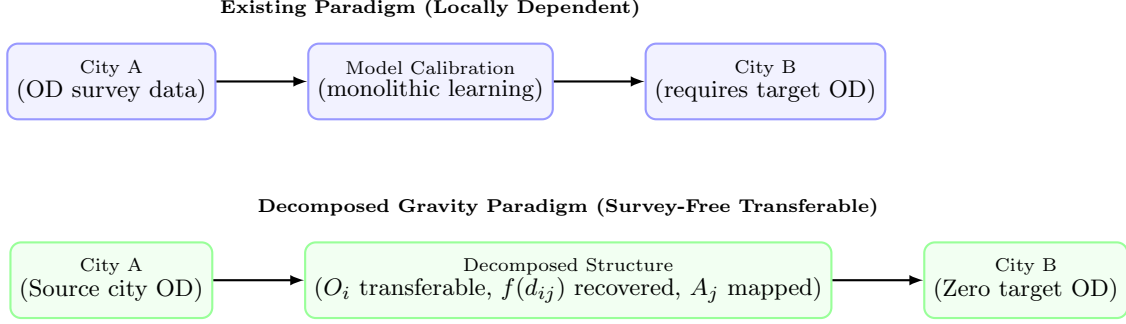
In this paper, we address this question by proposing a three-tier conceptual framework: (1) we identify the *transferable structural component* of spatial interaction as the city-specific distance-decay function, (2) we utilize the target-city aggregate trip-length distribution (TLD) as an *aggregate representation that reduces privacy exposure* that allows this transferable structural component

to be identified without revealing individual OD pairs, and (3) we apply this recovered decay in a *zero-shot calibration* setting to predict commuting flows. Under the primary **OD-free calibration setting**, the model recovers distance-decay parameters from target-city aggregate TLDs (e.g., Meta Movement Distribution Maps). In the **pure zero-shot fallback**, the model uses a national decay prior derived from source cities. We show that aggregate trip-length histograms contain sufficient information to recover the city-specific distance-decay structure with high fidelity, and our game-theoretic Shapley analysis attributes the vast majority of predictive capacity to distance decay within the investigated framework—consistent with the interpretation that distance decay functions as the primary transferable structural component of spatial interaction.

To investigate these relationships systematically, we organize the investigation around three research questions:

- RQ1 (Decay Recovery)** Can aggregate trip-length distributions recover the city-specific distance-decay structure without access to individual OD trajectories?
- RQ2 (Decay Impact)** How does distance decay impact zero-shot gravity transfer of commuting flows, and to what extent does its recovery from trip-length distributions drive transfer performance?
- RQ3 (Morphological Scope)** Under what urban morphological conditions does aggregate-calibrated decay perform most effectively, and what physical geography factors govern those conditions?

Our empirical investigation across 50 US metropolitan areas (25 source cities and 25 held-out target cities) evaluates these assumptions. The results indicate that distance decay represents a structurally transferable object across metropolitan areas, and that projecting spatial interactions onto the TLD provides a summary representation sufficient to identify this component. The resulting model achieves competitive performance across diverse urban morphologies, while remaining completely survey-free and privacy-compliant. The rest of this paper is organized around the three research questions (RQ1–RQ3) detailed above.



**Fig. 2** Conceptual comparison of the existing locally dependent modeling paradigm versus the decomposed gravity paradigm investigated in this study. Under the existing paradigm, models require target-city OD observations to retrain or calibrate parameters. In the decomposed paradigm, transferable origin production structures are learned from source cities, decay is recovered from target aggregate data, and attraction is mapped from local open features, enabling survey-free transfer without target-city OD surveys.

### 1.1 Classical and Structured Mobility Models

Gravity models (Wilson 1971; Tanner 1961) and radiation models (Simini et al. 2012) provide the basis of spatial interaction modeling. Gravity models employ physical metaphors to link flows to mass and distance in zones, while radiation models interpret the choice of destinations probabilistically on the basis of intervening opportunities. These models offer interpretability, but gravity models require local OD data to calibrate decay parameters, while radiation models underperform in cities with strong directional commuting preferences or constrained network topologies.

### 1.2 Deep Learning for OD Prediction

DeepGravity (Simini et al. 2021) uses neural networks to associate built-environment characteristics with flow volumes. Recent advancements include graph neural networks (Rong et al. 2023) and OSM-based commuter modeling (Atwal et al. 2025). These methods achieve strong predictive accuracy where training data is abundant, but require substantial target-city OD records for training, limiting their applicability in data-scarce environments.

### 1.3 Aggregate Mobility Data for Decay Calibration

Aggregated mobility datasets can be summarized to form trip-length distributions (TLDs)—marginal histograms of traveled distances—that

do not reveal any individual OD pairs (Papalardo et al. 2023). Previous studies suggest that such distributions match distance-decay models (Brockmann et al. 2006; Lenormand et al. 2016). However, there is no existing study on using such distributions to calibrate zero-shot cross-city transfer.

### 1.4 Cross-City Transfer Learning

Transfer learning across cities has been studied using similarity-based alignment (Wang et al. 2018), representation learning (Mendieta et al. 2023), and domain adaptation (Wang et al. 2022). However, existing methods require access to target-city observations or domain-specific calibration.

### 1.5 Research Gap and Positioning

Table ?? compares how existing model classes address the core challenge of zero-shot cross-city transfer. The table reveals three specific gaps: G1 (lack of aggregate-based decay calibration for zero-shot transfer), G2 (unquantified impact of distance decay on zero-shot transfer), and G3 (unknown urban morphological conditions for aggregate calibration). These gaps correspond directly to the research questions (RQ1–RQ3) evaluated in this study.

## 2 Problem Setting

We define three evaluation settings (Table 1) to analyze zero-shot cross-city transfer under different data constraints.

**Table 1** Operational and diagnostic evaluation settings.

| Setting / Control                  | Target City Mobility Data                 | Deployment / Analytical Purpose                      |
|------------------------------------|-------------------------------------------|------------------------------------------------------|
| Setting 1: Pure Zero-Shot Fallback | None (Open spatial features only)         | Fallback scenario under extreme data scarcity        |
| Setting 2: OD-Free Calibration     | Aggregate 1D TLD (no individual OD flows) | Primary proposed pipeline; enables local calibration |
| Oracle Diagnostic Control          | Target outflows $O_i$ controlled          | Diagnostic benchmark; isolates decay recovery        |

### 3 Methodological Design

**Core Idea.** This study structures cross-city transfer under a separable gravity framework:

$$T_{ij} = O_i \cdot \frac{A_j f(d_{ij})}{\sum_k A_k f(d_{ik})} \quad (1)$$

where flows are decomposed into origin production ( $O_i$ ), destination attraction ( $A_j$ ), and distance decay ( $f(d_{ij})$ ). We assume that: (1) distance decay is the dominant transferable component of spatial interaction (RQ2), (2) aggregate TLDs contain sufficient information to recover this decay structure without individual OD flows (RQ1), and (3) the recovered decay can be integrated with GBDT predicted outflows and open-data attraction heuristics to predict target-city flows (RQ3).

#### 3.1 Aggregate-Based Decay Calibration

We recover the distance-decay parameters  $\theta = (\gamma, \beta)$  from the aggregate trip-length distribution (TLD)  $p(d)$  — a marginal histogram of trip distances. Formally:

$$p(d) = \sum_{i,j} T_{ij} \cdot \mathbf{1}[d_{ij} \approx d] \quad (2)$$

Because  $p(d)$  is conditioned by zone geometries and employment opportunity arrangements, we formalize the parameter recovery using an exposure-corrected likelihood model. Let  $y_b$  be the observed commuter count in distance bin  $b \in \{1, \dots, K\}$  from the target-city TLD. Under the gravity framework, the predicted flow is:

$$\hat{T}_{ij}(\theta) = O_i \frac{A_j f_\theta(d_{ij})}{\sum_k A_k f_\theta(d_{ik})} \quad (3)$$

where  $O_i$  is target outflow,  $A_j$  is the attraction heuristic (Section 3.4), and  $f_\theta$  is the decay function. The predicted probability of a trip falling

into bin  $b$  is:

$$P_\theta(b) = \frac{\sum_{i,j: d_{ij} \in b} \hat{T}_{ij}(\theta)}{\sum_{i,j} \hat{T}_{ij}(\theta)} \quad (4)$$

This predicted probability  $P_\theta(b)$  explicitly corrects for spatial opportunity distribution and zone geometries. Assuming a multinomial distribution of trips across the distance bins, we estimate  $\theta = (\gamma, \beta)$  by maximizing the multinomial log-likelihood between the observed fractions  $\bar{y}_b = y_b / \sum_{b'} y_{b'}$  and predicted probabilities  $P_\theta(b)$ :

$$\mathcal{L}(\theta) = \sum_{b=1}^K \bar{y}_b \log P_\theta(b) \quad (5)$$

Minimizing  $-\mathcal{L}(\theta)$  recovers the parameters from aggregate data without individual zone-to-zone records. In production,  $O_i$  in Eq. (3) is replaced by the GBDT-predicted  $\hat{O}_i$ ; the model structure is shown in Fig. 3.

#### 3.2 Distance Decay Function

We use the Tanner multiplicative deterrence function (Tanner 1961), which combines power-law and exponential terms:

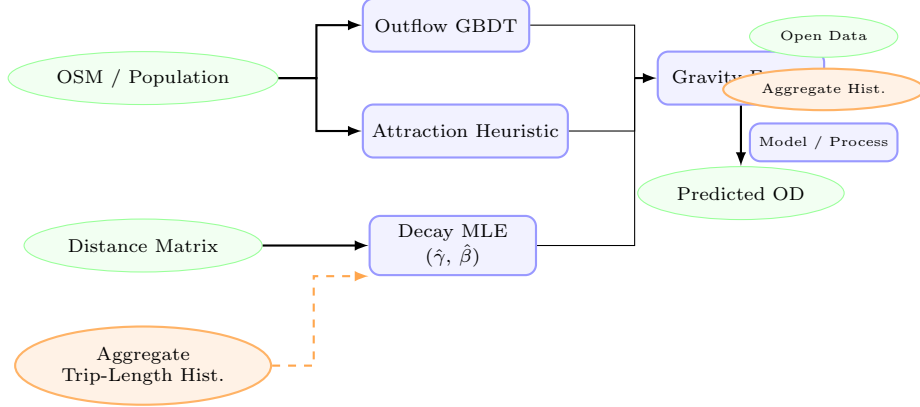
$$f(d_{ij}; \beta, \gamma, \delta) = (d_{ij} + \delta)^{-\gamma} \cdot \exp(-\beta d_{ij}) \quad (6)$$

where  $\beta > 0$  is the exponential decay rate governing long-range trip attenuation,  $\gamma \geq 0$  is the power-law exponent modeling short-to-medium range trip distribution, and  $\delta > 0$  is the adaptive geometric anchor.

The offset  $\delta$  prevents singularity at  $d_{ij} = 0$  and adapts to zone size:

$$\delta = 0.5 \times \bar{R}_{\text{zone}}, \quad \bar{R}_{\text{zone}} = \frac{1}{N} \sum_{i=1}^N \sqrt{\frac{\text{Area}_i}{\pi}} \quad (7)$$

where  $\bar{R}_{\text{zone}}$  is the mean zone radius. Parameters  $(\beta, \gamma)$  are estimated via MLE on the aggregate



**Fig. 3** Overview of the aggregate-calibrated gravity model structure. Globally available open-data features (green ellipses) are used to predict trip production ( $O_i$ ) via a GBDT model and to compute destination attraction ( $A_j$ ) via a training-free min-max heuristic. The **Aggregate Trip-Length Histogram** (orange ellipse, dashed arrow)—which provides aggregate summaries that significantly reduce privacy exposure compared with individual trajectories—feeds the Decay MLE block to recover city-specific parameters ( $\hat{\gamma}, \hat{\beta}$ ) without requiring any individual OD trajectories. The three components are combined via multiplicative gravity fusion to predict survey-free flow matrices ( $T_{ij}$ ).

distance-bin distribution. Calibration and optimization details are in the Appendix. Because only binned trip-length histograms are used—not individual OD records—this procedure is compatible with strict data-governance regulations (e.g., GDPR). For numerical implementation and computational efficiency, the theoretical offset  $\delta$  is regularized by clamping the minimum distance to a small epsilon ( $d_{\min} = 10^{-6}$ ), which yields mathematically equivalent results as  $d_{ij} \gg d_{\min}$  for all cross-zone flows ( $i \neq j$ ).

### 3.3 Origin Outflow Component ( $O_i$ )

The trip production  $O_i = \sum_j T_{ij}$  is the total trips originating from zone  $i$ . We predict  $O_i$  with a GBDT regressor trained on a parsimonious set of **6 core macroscopic features**: total population ( $P_i$ ), total POI count ( $\text{POI}_i$ ), zone area ( $\text{Area}_i$ ), population density ( $\rho_{\text{pop},i}$ ), POI density ( $\rho_{\text{poi},i}$ ), and road density ( $\text{rd}_i$ ). This parsimonious feature set is chosen based on dimensional scaling in urban physics and Occam’s razor, and avoids overfitting to source-domain geometries. Hyperparameters are in Appendix C.

### 3.4 Destination Attraction Component ( $A_j$ )

The destination attraction  $A_j$  represents a zone’s opportunity capacity. To avoid source-domain overfitting in zero-shot transfer, we define a

training-free, scale-invariant heuristic:

$$\hat{A}_j = \frac{1}{2} [\text{minmax}(P_j) + \text{minmax}(\text{POI}_j)] \quad (8)$$

where  $\text{minmax}(x_j) = (x_j - \min_k x_k) / (\max_k x_k - \min_k x_k + 10^{-9})$ ,  $P_j$  is the gridded population count, and  $\text{POI}_j$  is the OpenStreetMap POI count. This normalizes absolute scale differences while preserving local opportunity hierarchies. In Section ??, the ablation analysis (Table 3) validates that this training-free heuristic contributes meaningfully to zero-shot performance, with its removal causing a 4.4% CPC drop. The proposed formulation is empirically sufficient within the evaluated scope without introducing source-domain overfitting.

### 3.5 Algorithmic Flow

The complete pipeline for training and zero-shot deployment is formalized in Algorithm 1.

### 3.6 Data Sources

We use three globally available open data sources: built-environment spatial descriptors (OpenStreetMap POIs and road density), gridded demography (WorldPop), and aggregate movement telemetry maps (Meta Movement Distribution Maps (Meta 2026)). Ground-truth origin-destination (OD) flows between census tracts are obtained from a third-party human mobility

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**Algorithm 1: Aggregate Calibration and Zero-Shot Prediction**

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**Input:** Source cities data  $\{\mathbf{X}^{(c)}, \mathbf{T}^{(c)}\}_{c=1}^C$ ; Target city spatial features  $\mathbf{X}^{(t)}$ ; Target city aggregate trip-length histogram  $\{y_b\}_{b=1}^K$  (or national prior if unavailable)

**Output:** Predicted Target City OD Matrix  $\hat{\mathbf{T}}^{(t)}$

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**Phase 1: Offline Training (Source Cities)**

1. **Production Model Training:** Train the outflow model  $g(\mathbf{x})$  (GBDT) on pooled source-city spatial features  $\mathbf{X}^{(c)}$  and ground-truth outflows  $O_i^{(c)}$ .

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**Phase 2: Online Calibration & Zero-Shot Transfer (Target City)**

2. **Decay Calibration:** Recover target-city decay parameters  $(\hat{\beta}, \hat{\gamma})$  via MLE on the target city’s aggregate distance-bin histogram  $\{y_b\}_{b=1}^K$ . (If unavailable, fall back to the source-derived national prior  $(\beta_{\text{nat}}, \gamma_{\text{nat}})$ ).

3. **Attraction Mapping:** Compute target city zone attractions using the training-free heuristic:  $\hat{A}_j^{(t)} = \frac{1}{2} \left( \min_{\mathbf{x}} (P_j^{(t)}(\mathbf{x})) + \min_{\mathbf{x}} (\text{POI}_j^{(t)}(\mathbf{x})) \right)$ .

4. **Zero-Shot Target Prediction:**

a. Predict target city outflows:  $\hat{O}_i^{(t)} = g(\mathbf{x}_i^{(t)})$ .

b. Compute adaptive target offset:  $\delta^{(t)} = 0.5 \times \bar{R}_{\text{zone}}^{(t)}$ .

c. Fuse components and normalize outflows:  $\hat{T}_{ij}^{(t)} = \hat{O}_i^{(t)} \frac{\hat{A}_j^{(t)} (d_{ij} + \delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ij}}}{\sum_k \hat{A}_k^{(t)} (d_{ik} + \delta^{(t)})^{-\hat{\gamma}} e^{-\hat{\beta} d_{ik}}}$ .

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dataset covering 50 US metropolitan areas (25 source cities for GBDT production training and 25 held-out target cities for testing). The target OD matrices are kept completely blind to the model components and are used exclusively as ground truth for validation.

### 3.7 Meta MDM Harmonization

Meta MDM provides aggregate travel fractions at the county level grouped into 3 physical distance bins:  $[0, 10 \text{ km}]$ ,  $(10, 100 \text{ km}]$ , and  $> 100 \text{ km}$ . To harmonize these county-level travel fractions into tract geometries for calibration:

1. County-level trip-length distributions are averaged using WorldPop-derived population shares as weights:  $p_{\text{metro}}(b) = \sum_{c \in \mathcal{C}} w_c \cdot p_c(b)$ , where  $w_c$  is the population share of intersecting county  $c$ .
2. Tract pairs  $(i, j)$  are mapped to these 3 bins based on their centroid distances  $d_{ij}$ .
3. The predicted probability of a flow falling into Meta bin  $b$  is computed by summing and normalizing predicted tract-level flows in that bin:  $P_\theta(b) = \sum_{i,j: d_{ij} \in \text{Range}(b)} \hat{T}_{ij}(\theta) / \sum_{i,j} \hat{T}_{ij}(\theta)$ . This normalization bridges the continuous gravity formulation with Meta’s 3-bin telemetry.

### 3.8 Outflow Model Features

We predict origin outflow  $O_i$  using a parsimonious set of 6 core built-environment features: population, POI count, zone area, population density, POI density, and road density. These 6 features were selected from an initial 30-candidate set using a cross-city SHAP stability analysis, showing no performance loss compared to the full feature set.

### 3.9 Urban Morphology Grouping (RQ3)

To investigate how physical geography modulates distance decay’s role, the 25 target cities are classified into four morphological groups (Table 2) based on urban structure and geography (González et al. 2008).

## 4 Results

### 4.1 RQ1: Can aggregate distributions recover distance decay?

Crucially, evaluating downstream OD prediction under oracle outflow demonstrates that using the recovered parameters yields performance close to the ground truth parameters (Table ??), resulting in a negligible average gap of just 0.11%. This



**Table 2** Morphological classification of target cities.

| Morphology Group              | Cities                                                                                      |
|-------------------------------|---------------------------------------------------------------------------------------------|
| Dense / Transit ( $n = 5$ )   | Boston, Long Beach, New York, Philadelphia, Washington DC                                   |
| Sprawling / Grid ( $n = 10$ ) | Atlanta, Dallas, El Paso, Houston, Jacksonville, Las Vegas, Mesa, Omaha, San Antonio, Tulsa |
| Polycentric ( $n = 7$ )       | Albuquerque, Charlotte, Detroit, Louisville, Milwaukee, Raleigh, San Jose                   |
| Coastal ( $n = 3$ )           | Portland, Seattle, Virginia Beach                                                           |

suggests that parameter recovery successfully supports calibration in downstream flow prediction without requiring individual trajectories. Note that this comparison is conducted under oracle outflow, isolating the decay component; GBDT production estimation remains the primary bottleneck (see Table 4).

The parameter recovery is strong for  $\gamma$  but only moderate for  $\beta$  (Table ??). The weaker recovery of  $\beta$  likely reflects the sensitivity of the exponential term’s tail behavior to coarse histogram binning. Critically, the negligible downstream CPC gap implies that  $\beta$  recovery errors do not substantially distort the decay curve within typical commuting distances.

#### 4.1.1 Validation via Real-World Meta Mobility Prior (RQ1)

To verify the practical feasibility of using tech-platform telemetry, we evaluate the decay parameters recovered from Meta’s actual Movement Distribution Maps prior (**Meta MDM Prior (3-Bin)**) on the 25 target cities. We compare its downstream CPC performance under oracle outflows against models calibrated on third-party databases aggregated into 3 bins (**Third-Party Baseline (3-Bin)**) and 20 equal-width bins (**Third-Party Oracle (20-Bin)**), as well as the zero-shot DeepGravity baseline.

As shown in Table ??, the survey-free **Meta MDM Prior (3-Bin)** calibration yields a minor performance gap (under 5.0%) compared to the survey oracle. When compared to the corresponding 3-bin baseline, the gap narrows to under 4.0%.

This suggests that publicly released location fractions serve as a viable calibration signal in place of surveys.

To evaluate robustness under extreme data sparsity where target TLDs are missing, we evaluate the National Decay Fallback (Setting 1). The national prior parameters are derived as the median of estimates fitted on source cities. This fallback yields stable performance under the Oracle Outflow control, strongly outperforming the Uniform baseline (which completely omits distance decay and serves as a structural lower bound). When destination attractiveness  $A_j$  is integrated, the full Oracle National Prior model recovers most of the survey oracle performance. In the fully deployed GBDT outflow setting, this fallback introduces a minor error propagation loss (6–7%), confirming the viability of zero-shot transfer without target mobility data.

The performance of the localized Meta MDM calibration is sensitive to local outlier dynamics. Specifically, Atlanta serves as an extreme outlier, exhibiting a localized drop compared to the 3-bin baseline. This drop is driven by the sprawling polycentric layout of the city coupled with telemetry binning noise. The detailed case study of this outlier behavior is provided in Appendix A.2.

#### 4.1.2 Decay Bin Sensitivity Analysis (RQ1)

Sweeping the distance bins  $K \in \{3, 5, 10, 20\}$  shows that downstream CPC stabilizes at  $K \geq 5$  (compared to the full OD ground truth of 0.7411, Table ??). This indicates that while extremely coarse  $K = 3$  binning suffers from a modest information loss (3.2% relative gap),  $K \geq 5$  bins is

empirically sufficient to capture the city-specific decay profile with negligible degradation. Full sensitivity details and the recovery error heatmap are in Appendix A.2.

## 4.2 RQ2: Why is recovery sufficient?

To address RQ2, we first evaluate the origin production component ( $O_i$ ) in isolation to validate the predictive quality of GBDT outflows, and then implement a game-theoretic Shapley value analysis to quantify the relative contributions of the model components.

### 4.2.1 Origin Production Transferability

To isolate the contribution of  $O_i$ , we hold  $A_j$  (OSM heuristic) and  $(\gamma, \beta)$  (national prior) fixed, then compare three outflow strategies across 25 target cities: a naive population baseline ( $O_i \propto \text{Population}_i$ ), GBDT-predicted outflows, and oracle outflows.

We find that GBDT outflows (CPC 0.6896) beat the naive population baseline (0.6758) and achieve 93.2% of the oracle upper bound (CPC 0.7398) under zero-shot transfer. Robustness experiments are in Appendix B.

**Feature Selection via SHAP Stability Analysis.** Evaluating GBDT model performance across nested subsets of top SHAP features indicates that zero-shot prediction performance plateaus at  $K \approx 10$  features. The detailed feature taxonomy, SHAP stability criteria, and feature count sensitivity are provided in Appendix C.

### 4.2.2 Factorial Ablation and Shapley Value Analysis

To evaluate the relative structural importance of each gravity component for zero-shot transfer (RQ2), we conduct a factorial ablation across the 25 target cities. Each component is disabled in turn—replaced by a uniform-weight fallback—to isolate its marginal contribution under our gravity equations.

We implement a game-theoretic analysis using Shapley values (Shapley 1953) across all  $2^3 = 8$  component coalitions. Over the uniform flow baseline (CPC 0.4263), the Shapley values attribute a mean CPC gain of 0.2258 (85.8% of total gain) to distance decay  $f(d)$ , 0.0208 (7.9% of total gain) to

attractiveness  $A_j$ , and 0.0167 (6.4% of total gain) to outflow production  $O_i$ . Note that the factorial ablation baseline in Table 3 is the full model using the national decay prior (CPC 0.6896), where  $f(d)$  removal yields a CPC of 0.4489.

This indicates that distance decay is the structurally dominant component in predictive performance within the investigated framework, explaining why recovery of the decay parameters (RQ1) is a primary determinant of spatial flow recovery.

## 4.3 RQ3: Does aggregate-calibrated decay provide a practically deployable alternative to survey-based calibration?

We evaluate the proposed aggregate-calibrated decay model against zero-shot and locally calibrated baselines under fully survey-free settings (RQ3). The results are summarized in Table 4.

Oracle outflows do not materially affect DeepGravity because it jointly models production and destination choice. Conversely, the proposed gravity model matches the locally-fit baseline under oracle outflows.

Applying the Wilcoxon signed-rank test across the target cities, the difference between the proposed model (Oracle  $O_i$ , Meta MDM, CPC 0.7337) and the locally-calibrated Traditional Tanner gravity baseline (CPC 0.7382) is statistically significant but practically negligible (raw  $p = 0.00737$ ). A Two One-Sided Test (TOST) formally establishes that the proposed model is equivalent to the locally-fit Tanner baseline within a tight margin of  $\Delta_{\text{eq}} = 0.01$  CPC ( $p = 0.00326$ ). The proposed model significantly outperforms the Radiation Model baseline (Wilcoxon Bonferroni-adjusted  $p < 10^{-6}$ ).

### 4.3.1 Exploratory Urban Morphology Analysis (RQ3)

To address RQ3, we analyze model performance across the morphological groups (Table 5).

The structured physical formulation exhibits its strongest performance in Coastal environments (CPC 0.714). The performance gap between the physics-constrained gravity model and the neural DeepGravity model narrows to just 5.8% in Coastal cities compared to a wider 10.3% gap in Dense / Transit cities. This morphological



**Table 3** Factorial Ablation and Shapley Value Contributions (Average across  $N = 25$  held-out target cities). The full-model baseline (top row) uses a fixed national decay prior  $^{\ddagger}$  to isolate component contributions.

| $O_i$ | $f(d)$          | $A_j$ | CPC    | Factorial Contribution                          | Shapley Value  |
|-------|-----------------|-------|--------|-------------------------------------------------|----------------|
| ✓     | ✓ $^{\ddagger}$ | ✓     | 0.6896 | Baseline (all components, national prior decay) | —              |
| ×     | ✓ $^{\ddagger}$ | ✓     | 0.6645 | $O_i$ removal: 3.6% loss                        | 0.0167 (6.4%)  |
| ✓     | ×               | ✓     | 0.4489 | $f(d)$ removal: 34.9% loss                      | 0.2258 (85.8%) |
| ✓     | ✓ $^{\ddagger}$ | ×     | 0.6574 | $A_j$ removal: 4.7% loss                        | 0.0208 (7.9%)  |

$^{\ddagger}$ In all ablation rows,  $f(d)$  is frozen at the national prior ( $\gamma_{\text{nat}}, \beta_{\text{nat}}$ ) to isolate the contribution of  $O_i$  and  $A_j$ . The full pipeline uses aggregate-recovery.

**Table 4** Multi-Baseline Comparison (25 Held-Out Target Cities)

| Model                                                                                  | Mean CPC | Std Dev | Type           | Features |
|----------------------------------------------------------------------------------------|----------|---------|----------------|----------|
| <b>Group 1: Survey-Free Transfer (No target OD data)</b>                               |          |         |                |          |
| Proposed Model (Survey-Free, Meta MDM)                                                 | 0.6860   | 0.0391  | Decomposed     | 6        |
| Proposed Model (Survey-Free, National Fallback)                                        | 0.6896   | 0.0384  | Decomposed     | 6        |
| DeepGravity (Zero-Shot)                                                                | 0.7529   | 0.0285  | Neural E2E     | 27       |
| <b>Group 2: Oracle Outflow Transfer (True outflows <math>O_i^{\text{true}}</math>)</b> |          |         |                |          |
| Proposed Model (Oracle $O_i$ , Meta MDM)                                               | 0.7337   | 0.0256  | Decomposed     | 6        |
| Proposed Model (Oracle $O_i$ , National Prior)                                         | 0.7398   | 0.0248  | Decomposed     | 6        |
| DeepGravity (Oracle $O_i$ Zero-Shot)                                                   | 0.7526   | 0.0283  | Neural E2E     | 27       |
| <b>Group 3: Locally-Calibrated Baselines (Fitted on target cities)</b>                 |          |         |                |          |
| Naive Population Baseline                                                              | 0.6758   | 0.0384  | Baseline       | 0        |
| Traditional Tanner Gravity                                                             | 0.7382   | 0.0322  | Structure-only | 0        |
| Traditional Exponential Gravity                                                        | 0.6700   | 0.0350  | Structure-only | 0        |
| Radiation Model                                                                        | 0.4851   | 0.0516  | Param-free     | 0        |

pattern is tentatively consistent with the interpretation that physical distance constraints amplify the advantage of structured decay-based models, though given the small group sizes, these patterns should be treated as preliminary.

To establish a continuous benchmark, we analyze the correlation between the performance gap ( $\Delta\text{CPC} = \text{CPC}_{\text{DeepGravity}} - \text{CPC}_{\text{Proposed}}$ ) and density. The performance gap is positively correlated with median employment density ( $r = 0.263$ ) and population density ( $r = 0.252$ ). This indicates that in dense, transit-oriented environments, multi-modal networks decouple flows from geometric distance, allowing neural models to exploit non-linear feature interactions, whereas in lower-density or physically constrained environments, interactions are governed by distance.

In summary, for RQ3, the aggregate-calibrated model enables survey-free zero-shot deployment

that is statistically equivalent to locally-fit models in lower-density or physically constrained urban areas.

## 5 Discussion

The three research questions (RQ1–RQ3) defined in Section 1 correspond directly to the three gaps (G1–G3) identified in Table ??: G1 addressed by demonstrating aggregate-based decay recovery (Section 4.1), G2 by quantifying decay dominance via Shapley analysis (Section 4.2), and G3 by the exploratory morphological evaluation (Section 4.3). The following discussion distinguishes rigorously established empirical findings from claims that warrant further cross-regional verification.

**Table 5** Survey-Free Performance by Urban Morphology (25 Held-Out Target Cities)

| Morphology Type  | # Cities | Proposed CPC | DeepGravity CPC | Rel. Gap (%) |
|------------------|----------|--------------|-----------------|--------------|
| Dense / Transit  | 5        | 0.651        | 0.726           | 10.3%        |
| Sprawling / Grid | 10       | 0.691        | 0.760           | 9.0%         |
| Polycentric      | 7        | 0.704        | 0.759           | 7.2%         |
| Coastal          | 3        | 0.714        | 0.758           | 5.8%         |

### 5.1 Empirical Evidence (Established Findings)

Our empirical results across 50 US metropolitan areas (25 source cities for training and 25 held-out target cities for evaluation) provide direct statistical evidence for the following three claims within the evaluated scope:

- 1. High-Fidelity Decay Parameter Recovery:** We empirically demonstrate that aggregate trip-length histograms (serving as an aggregate representation that reduces privacy exposure) preserve sufficient information to estimate per-city distance decay curves. The recovered exponents correlate strongly ( $r = 0.909$  for the power-law exponent  $\gamma$ ) with those fitted on individual-level OD flows.
- 2. TOST Equivalence under Oracle Outflows:** When target production totals are controlled (Oracle Diagnostic Control), we formally establish via a Two One-Sided Test (TOST) that the aggregate-calibrated gravity model is equivalent to the locally-calibrated Tanner gravity model within a tight margin of  $\Delta_{\text{eq}} = 0.01$  CPC (corresponding to a one-percentage-point tolerance in the CPC similarity index,  $p = 0.00326$ ).
- 3. Decay Dominance within the Decomposed Gravity Game:** Using game-theoretic Shapley values, the results support the finding that distance decay is the dominant contributor to zero-shot prediction capacity (85.8% of CPC gain) specifically within our separable gravity framework, providing a mathematical justification for why local decay calibration is the critical transferable structural component for zero-shot transfer.

The empirical recovery performance supports—but does not mathematically prove—the separability approximation adopted in this study.

### 5.2 Conceptual Intuition: Marginal Projection under Separability

To understand why aggregate trip-length distributions contain sufficient information to recover city-specific distance-decay parameters, it is helpful to examine the problem from a probabilistic perspective. The full origin-destination (OD) matrix represents a discrete joint probability distribution  $P(X, Y)$  of trips over a two-dimensional spatial grid of origins  $X$  and destinations  $Y$ . The trip-length distribution (TLD) represents the projection of this joint distribution onto a one-dimensional distance domain  $D$ . Formally, the TLD is the marginal distribution of travel distances:

$$P(D = d) = \sum_{x \in X} \sum_{y \in Y} P(X = x, Y = y) \cdot \mathbf{1}[d_{xy} \approx d] \quad (9)$$

Under the proposed decomposed gravity hypothesis, the joint probability is structurally constrained. The flow between any zone pair is separable into independent factors representing origin production capacity, destination attraction capacity, and distance deterrence. Because the distance decay function  $f(d)$  is the only component in the joint model that depends explicitly on the spatial distance  $d_{xy}$ , the projection of  $P(X, Y)$  onto the distance dimension  $D$  preserves the shape of  $f(d)$ .

While the spatial distribution of opportunities (the arrangement of population masses  $O_i$  and POI centers  $A_j$ ) acts as a geometric filter that distorts the raw marginal distribution  $P(D)$ , this distortion is purely a function of the known built environment. The exposure-corrected MLE (Equation 5) acts as a deconvolution mechanism: it utilizes the target city’s known spatial configurations to filter out the geometric exposure bias, thereby isolating the latent distance-decay parameters. In other words, because the decay behavior is assumed to be a uniform, city-wide

spatial friction, it manifests consistently across all zone pairs, allowing its parametric shape to be reconstructed from the marginal distance distribution once local exposure is controlled. Aggregate trip-length distributions are sufficient for recovering the decay parameters conditional on known spatial opportunity distributions and the gravity formulation.

This probabilistic deconvolution also explains the performance comparison observed when transitioning from oracle outflows to GBDT-predicted outflows (Setting 2 vs. Setting 1). Under oracle outflows evaluated over all 25 target cities, the ground-truth-derived national prior slightly outperforms the localized Meta MDM calibration (CPC 0.7398 vs. 0.7337, a gap of 0.0061 CPC), because the national prior incorporates clean, unbinned historical survey records while Meta MDM is restricted to a privacy-masked, 3-bin telemetry configuration. When transitioning to GBDT-predicted outflows ( $\hat{O}_i$ ), the national prior’s advantage narrows to just 0.0036 CPC (CPC 0.6896 vs. 0.6860). This behavior is driven by the interaction between data source quality and error propagation: the continuous national prior acts as a regularizer during gravity normalization (Equation 1), smoothing out localized flow fluctuations and reducing sensitivity to production prediction noise, whereas the binned Meta MDM decay lacks this continuous smoothing and propagates production errors.

Critically, the fact that the survey-free Meta MDM calibration matches the ground-truth-derived national prior within 0.0036 CPC under GBDT outflows (and within 0.0061 CPC under oracle outflows) demonstrates that privacy-compliant, platform-sourced telemetry can serve as an exceptionally close surrogate without causing any substantial degradation in forecasting accuracy. While a high-quality national prior is a powerful option when target cities share regional similarities with the training database, such a prior cannot be constructed in data-scarce regions or developing nations lacking historical travel surveys. In these zero-data deployment scenarios, the global availability of Meta MDM provides an essential, training-free calibration signal that represents the only viable alternative to costly surveys, making it a critical deployment pathway for international generalizability with negligible performance loss.

### 5.3 Hypotheses Requiring Further Verification

While our empirical evidence is robust, several key claims remain hypotheses that require further cross-regional verification:

1. **Global Transferability:** Whether the learned GBDT production weights and the open-data attraction heuristic generalize outside the United States (e.g., to dense Asian cities or informal settlements in developing countries) remains a hypothesis.
2. **Physical Causality:** We show that distance decay functions as the central explanatory driver of predictive performance. However, this statistical association does not prove that geometric distance causally determines travel behavior. In reality, distance is a proxy for travel time, transport cost, and individual behavioral constraints.
3. **Separability Approximation:** The assumption that spatial interaction components (production, attraction, decay) are strictly separable and independent remains a structural approximation. In highly integrated transit networks, these components likely exhibit non-linear coupling.

### 5.4 Practical Deployment Implications

By demonstrating that high-fidelity decay calibration is achievable using only target-city aggregate histograms, this study enables a new paradigm for deploying mobility models. Historically, calibrating per-city gravity models required target household travel surveys or invasive individual-level GPS telemetry, both of which raise significant privacy concerns and are restricted under frameworks like GDPR.

By leveraging globally available open-data features (OSM, WorldPop) and target-city aggregate representations (like the Meta Movement Distribution Maps), planning agencies can calibrate and deploy structured, interpretable gravity models in data-scarce or highly regulated environments. This bypasses the need for proprietary survey collection while maintaining strict data firewalls.

## 6 Threats to Validity

To ensure the objective interpretation of our findings, we systematically categorize the threats to the validity of our transfer learning workflow:

**Not Pure Zero-Shot)** The primary proposed calibration setting (Setting 2) relies on target-city aggregate TLDs to calibrate parameters. While this bypasses individual zone-to-zone OD surveys, it still requires aggregate target-city travel data. Although we evaluate a pure zero-shot fallback (Setting 1) requiring zero target mobility data, under deployed GBDT outflows, Setting 1 actually outperforms Setting 2 slightly (CPC 0.6896 vs 0.6860), suggesting that localized calibration from highly binned target mobility telemetry does not necessarily yield downstream gains compared to a robust national fallback. Under oracle outflows, the full Oracle National Prior model achieves CPC 0.7398 (retaining 99.9% of the 20-bin survey oracle) compared to CPC 0.7337 under the Oracle Meta MDM prior on the 25-city full split.

**Overestimation Risk** In Experiment A, the aggregate histograms used for the oracle comparison are synthetically compiled from the target ground-truth OD matrix. While useful as a diagnostic bound, this does not capture data alignment mismatches of real-world deployment. When switching to independently sourced Meta MDM data, the oracle outflow CPC drops from 0.7403 (synthetic 20-bin oracle) to 0.7080 (Meta MDM 3-bin), a gap of  $-4.36\%$ , suggesting that synthetic aggregation overestimates real-world calibration quality due to different bin configurations and data sources.

1. **Morphology Sample Constraints (Small-n):** The morphological classification of target cities is manually defined, and some categories have small sample sizes (Coastal,  $n = 3$ ). While the performance differences align with urban geography theories, they are exploratory and lack the statistical power required for generalized claims.
2. **Spatial Error Blindness:** The Common Part of Commuters (CPC) is a global similarity index. It evaluates overall matrix overlap but is blind to localized spatial error structures, such as systematic prediction errors along specific high-volume transit corridors.

3. **Euclidean Distance Simplification:** We proxy travel distance using Euclidean distance between zone centroids. This ignores network routing circuitry, travel time, and multi-modal transit friction, which decouple travel from geometric distance in dense transit environments.
4. **Geographic Generalizability:** The empirical evaluation is restricted to US cities. Commuting patterns, urban density profiles, and land-use arrangements in these cities may not represent those in Europe, Asia, or South America.

## 7 Conclusions

This study investigated the zero-shot cross-city transfer problem through a three-tier conceptual architecture, moving away from ad-hoc calibration toward a structural understanding of urban spatial interaction. By focusing on 50 US metropolitan areas (25 source cities for training and 25 held-out target cities for evaluation), we established that: (1) the city-specific distance-decay function represents the primary transferable structural component of spatial flow, (2) the aggregate trip-length distribution acts as an aggregate representation that reduces privacy exposure sufficient to identify this component, and (3) this relationship can be successfully leveraged for zero-shot gravity model calibration. Our work leads to three core conclusions:

### 7.1 What Did We Discover?

Empirically, we demonstrated that the target-city aggregate trip-length distribution (TLD)—even when heavily compressed into a small number of physical distance bins (e.g., Meta Movement Distribution Maps)—contains sufficient signal to identify the parameters of the distance-decay function. Under controlled oracle outflows, calibration on this aggregate measurement yields downstream prediction performance statistically equivalent to models calibrated directly on target-city individual-level OD flows. Mechanistically, our game-theoretic Shapley value analysis showed that distance decay dominates predictive capacity within this decomposed gravity framework, confirming its role as the primary transferable structural component. However, comparative evaluation also revealed that localized calibration from

real-world telemetry does not significantly outperform a robust national decay fallback. Despite this, while the national prior requires historical source data from cities with similar mobility characteristics, Meta MDM enables zero-shot deployment in data-scarce regions where such priors cannot be constructed, indicating that platform-sourced telemetry serves as an essential fallback pathway for geographic generalizability.

## 7.2 Why Does It Matter?

These findings matter because they redefine how spatial interaction models can be calibrated in data-scarce and privacy-regulated environments. Historically, identifying city-specific travel friction required highly sensitive individual-level surveys or proprietary cellular trajectories. By demonstrating that the primary transferable structural component (distance decay) can be robustly identified using a one-dimensional aggregate representation that reduces privacy exposure (the TLD), we provide urban planners with a deployable, survey-free forecasting pipeline. This architecture enables the calibration of structured, interpretable gravity models using only globally available open spatial data (OSM, WorldPop) and privacy-compliant aggregate indicators, bypassing GDPR constraints and reducing survey costs.

## 7.3 What Remains Unknown?

Despite these contributions, several structural elements of the urban mobility process remain unresolved:

- **Limits of the Separability Approximation:** While component decomposition (separating production, attraction, and decay) enables transferability, these components are likely non-linearly coupled in reality. How to model this coupled interaction while preserving zero-shot transferability remains unknown.
- **Generalizability Beyond US Geographies:** The empirical evidence is restricted to US metropolitan areas characterized by high private vehicle usage and specific density gradients. The transferability of these models to highly dense, multi-modal transit networks (e.g., in European or East Asian megacities) or informal settlements in the Global South requires further cross-regional verification.

- **Network Routing and Time-Distance**

**Friction:** Centroid-to-centroid Euclidean distance was used as a proxy for travel friction. Incorporating network routing, congestion-induced travel times, and multi-modal transit friction into the aggregate calibration process remains an open direction for future work.

## Appendix: Supplementary Materials

### Appendix A. Robustness of Aggregate-Based Distance-Decay Recovery

#### A.1 Estimation Details

To solve the maximum likelihood estimation (MLE) problem for target-city decay parameters  $\theta = (\gamma, \beta)$  without disaggregate flows, we adopt a robust and physically grounded initialization and optimization workflow based on closed-form moment estimates.

Good parameter initialization is critical for optimization convergence. For the exponential decay rate  $\beta$ , we utilize a closed-form moment estimate derived under the assumption of pure exponential deterrence and a uniform distribution of spatial opportunities. In this theoretical limit, the expected travel distance satisfies  $\mathbb{E}[d] = 1/\beta$ . We estimate the empirical mean trip distance  $\bar{d}_{\text{obs}}$  from the observed distance-bin histogram:

$$\bar{d}_{\text{obs}} = \sum_{k=1}^K b_k \cdot m_k \quad (10)$$

where  $b_k$  is the observed proportion of trips in bin  $k$ , and  $m_k$  is the midpoint of distance bin  $k$  (for the open-ended final bin, we use 1.5 times the lower boundary as a surrogate midpoint). The initial value  $\beta^{(0)}$  is set as:

$$\beta^{(0)} = \frac{1}{\bar{d}_{\text{obs}}} \quad (11)$$

For the power-law exponent  $\gamma$ , the initial value is set statically to  $\gamma^{(0)} = 1.0$ , which corresponds to the center of the empirically observed range [0.5, 2.0] in classical spatial interaction modeling literature.

The positivity constraints  $\beta > 0$  and  $\gamma \geq 0$  are strictly enforced during optimization via a parameter transformation. We define log-transformed parameters:

$$\tilde{\beta} = \log \beta, \quad \tilde{\gamma} = \log \gamma \quad (12)$$

and optimize the objective in the unconstrained log-transformed space  $\theta_{\log} = (\tilde{\gamma}, \tilde{\beta}) \in \mathbb{R}^2$ .

To guard against convergence to local optima due to the potential non-convexity of the exposure-corrected likelihood surface (especially under highly binned or noisy data), we run the L-BFGS-B optimization algorithm from 10 random restarts. The first run is initialized at the theoretical values  $\theta_{\log}^{(0)} = (\log \gamma^{(0)}, \log \beta^{(0)})$ . The remaining 9 runs are initialized by perturbing this central starting point with Gaussian noise:

$$\tilde{\gamma}_{\text{start}}^{(r)} \sim \mathcal{N}(\log \gamma^{(0)}, 0.25), \quad \tilde{\beta}_{\text{start}}^{(r)} \sim \mathcal{N}(\log \beta^{(0)}, 0.25) \quad (13)$$

The optimization is executed for each restart run, and the parameter set yielding the highest likelihood value  $\mathcal{L}(\theta)$  (Equation 5) is recovered via exponential back-transformation:  $\hat{\gamma} = \exp(\tilde{\gamma}^*)$  and  $\hat{\beta} = \exp(\tilde{\beta}^*)$ .

## A.2 Robustness Analysis

We evaluate the robustness of our MLE parameter recovery under non-ideal data scenarios:

*Bin Sensitivity Analysis* Sweeping the distance bins  $K \in \{3, 5, 10, 20\}$  shows that downstream prediction performance stabilizes at  $K \geq 5$ . Specifically, the mean CPC across target cities is 0.7174 for  $K = 3$ , 0.7346 for  $K = 5$ , 0.7394 for  $K = 10$ , and 0.7403 for  $K = 20$  (compared to the full OD ground truth baseline of 0.7411).

*Atlanta Case Study Outlier* In Section 6.2.1, Atlanta is identified as a localized outlier: its Meta MDM 3-bin CPC under oracle outflows is 0.5761 compared to 0.7334 under the Third-Party 3-bin baseline—a localized drop of 15.7 percentage points. This drop is driven by the spatial interaction between Atlanta’s sprawling polycentric layout and the telemetry binning noise in the Meta Location Fraction data. When Atlanta is removed, the target CPC for Meta MDM rises significantly from 0.7080 to 0.7135, retaining 96.4% of the corresponding Third-Party 3-bin baseline of 0.740 evaluated over the same 24-city subset.

*Missing Attractiveness Data ( $A_j$ )* Randomly removing 10%, 20%, and 50% of target-city destination attractions  $A_j$  yields highly stable decay recovery (e.g.,  $\gamma_{\text{MAE}} = 0.0395$  under 20% missing data vs. 0.0347 baseline).

*CBD Collapse (Extreme  $A_j$  Distortion)*

Disrupting the attraction  $A_j$  of the top 10% highest attractiveness zones (CBD collapse) still allows parameter recovery: replacing affected zones with the city-wide mean attraction ( $\gamma_{\text{mae}} = 0.0441$ ) and scaling them down by 80% ( $\gamma_{\text{mae}} = 0.0486$ ).

## A.3 Preprocessing and Normalization

Data preprocessing details are summarized in Table 6.

## Appendix B. Robustness of Origin Production Transfer

We analyze the sensitivity of zero-shot production transfer across two dimensions: training-set scale and prediction noise.

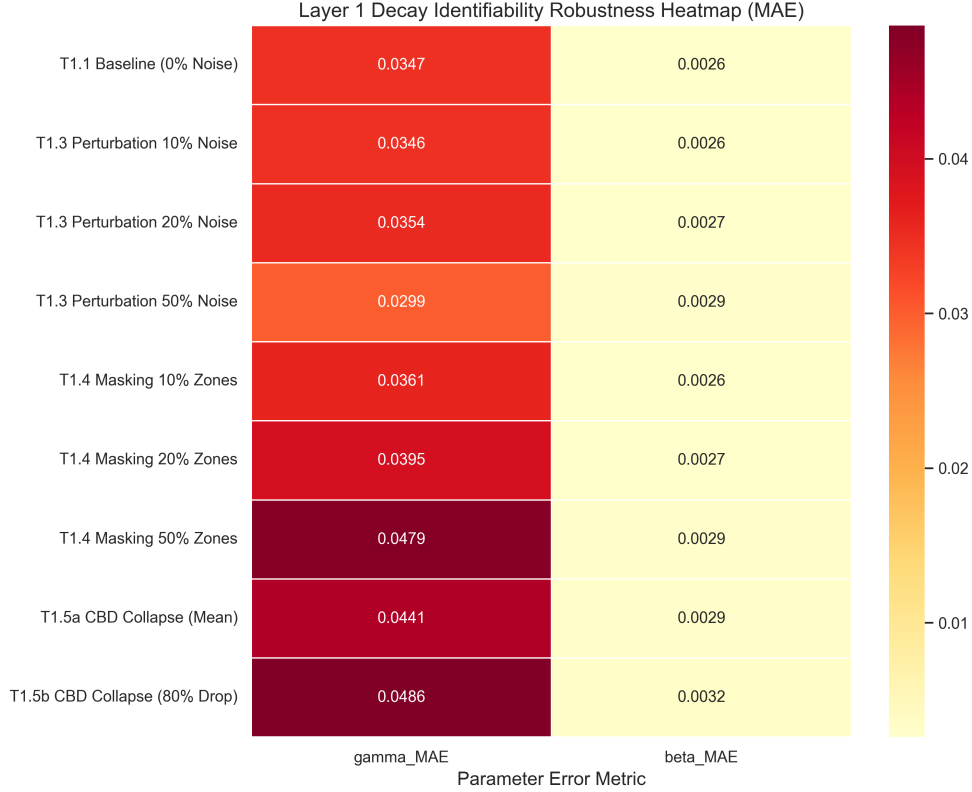
### B.1 Source-Scale Sensitivity

We analyze how the number of training environments affects zero-shot predictability by retraining the GBDT on subsets of  $n_{\text{src}} \in \{5, 10, 15, 20, 25\}$  source cities. Target-city evaluation shows that zero-shot prediction performance rises steeply from  $n_{\text{src}} = 5$  and saturates around 15–20 source cities. This indicates that only a modest number of source cities is required to learn stable, generalizable outflow characteristics.

### B.2 Noise Sensitivity

To evaluate the stability of origin transfer under prediction errors, we inject additive Gaussian noise at log-scale ( $\sigma \in \{10\%, 20\%, 40\%\}$ ) directly onto the predicted outflows  $\log \hat{O}_i$ . Under substantial injected noise (40%), the target CPC degrades from 0.6896 to 0.6626. While this  $-3.9\%$  drop is larger than the isolated production component contribution (3.6% Shapley value), it demonstrates that even highly corrupted outflow inputs do not collapse the spatial interaction matrix, because the production-constraint normalization restricts the error propagation.





**Fig. 4** Robustness of Aggregate-Based Distance-Decay Recovery (MAE) under varying levels of noise, missing data, and CBD collapse scenarios across all 25 held-out target cities.

**Table 6** Preprocessing and normalization steps applied to spatial features prior to model learning.

| Step            | Treatment              | Rationale                                    |
|-----------------|------------------------|----------------------------------------------|
| Missing Data    | Median imputation      | Rare (< 2%); resolved from spatial neighbors |
| Log Transform   | Population, POI counts | Stabilize right-skewed distributions         |
| Standardization | Per-city (not global)  | Preserve relative spatial hierarchy          |
| Outliers        | Retain                 | Important spatial hubs (CBDs); robust models |

## Appendix C. Feature Selection Sensitivity

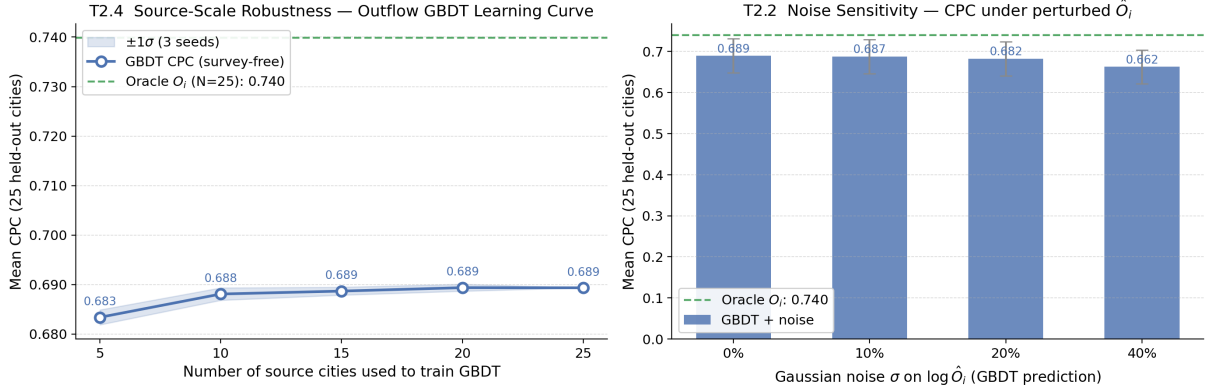
### C.1 Feature Taxonomy and Candidate Set (30 dimensions)

We initially construct 30 candidate features per zone. This candidate space comprises 23 scale and density variables (zone area, road length, residential population, total POI counts, 16 POI categories, and 3 density metrics: population density, POI density, and road density) and 7 city-wide Z-score context features. The feature taxonomy is shown in Fig. 6.

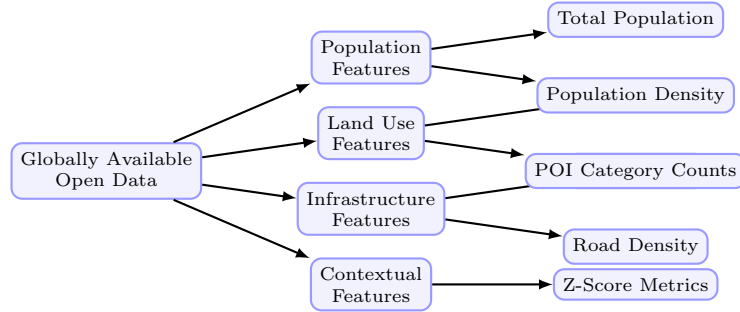
### C.2 SHAP-Based Feature Stability Selection

To identify features that generalize across cities, we apply SHAP importance analysis (Lundberg and Lee 2017) to the GBDT production model. Features are ranked by a cross-city stability score  $S_p = \mu_p / (\sigma_p + \epsilon)$ , where  $\mu_p$  and  $\sigma_p$  denote the mean and standard deviation of per-city GBDT SHAP importances and  $\epsilon = 10^{-5}$ . A feature is retained if  $\mu_p \geq 0.005$  and  $S_p \geq 1.0$ , ensuring both relevance and cross-city consistency. This analysis identifies 23 stable features from the 30-candidate

## Layer 2: Outflow ( $O_i$ ) Estimation — Bottleneck Analysis



**Fig. 5** Outflow bottleneck analysis. **Left:** CPC vs. training source cities ( $n_{src}$ ), showing performance saturation at 15–20 cities. **Right:** Noise sensitivity of CPC under additive Gaussian noise  $\sigma$  injected on predicted outflows  $\log \hat{O}_i$ .



**Fig. 6** Taxonomy of the globally available open data features used in the outflow model.

space (23.3% dimensionality reduction) and confirms that the 6 core structural features used in the deployed model (Section 3.3) are among the primary and most stable predictors, with no accuracy loss relative to the full candidate set.

### C.3 Feature Count Sensitivity

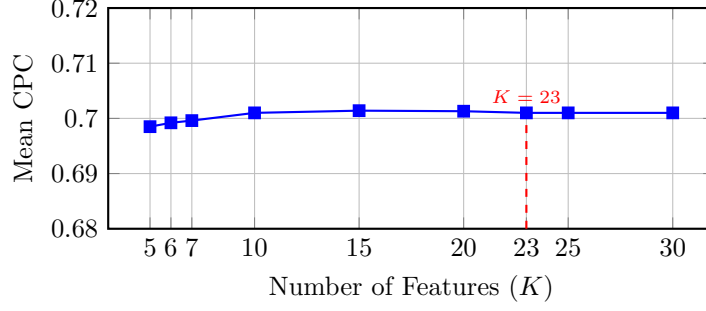
Evaluating the GBDT model performance across nested subsets of top SHAP features ( $K \in \{5, 6, 7, 10, 15, 20, 23, 25, 30\}$ ) shows that predictive performance plateaus between 10 and 23 features (CPC: 0.7010). This indicates that most transferable spatial information is captured by a compact set of built-environment indicators, with additional features providing marginal utility.

## Appendix D. List of Abbreviations

The key abbreviations and acronyms used throughout this paper are summarized and defined in Table 8.

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**Fig. 7** SHAP feature count sweep over the 30-candidate feature set: zero-shot CPC first reaches 0.7010 at  $K = 10$  and stabilizes in the range 0.7010–0.7014 through  $K = 23$ .

**Table 7** GBDT Model Hyperparameter Configurations

| Hyperparameter           | Outflow Model ( $O_i$ )            | Attraction Model ( $A_j$ , ablation only) |
|--------------------------|------------------------------------|-------------------------------------------|
| Base Learner             | Decision Tree                      | Decision Tree                             |
| Loss Function            | Least Squares (MSE on $\log O_i$ ) | Least Squares (MSE on $A_j$ )             |
| Learning Rate ( $\eta$ ) | 0.05                               | 0.05                                      |
| Number of Estimators     | 150                                | 200                                       |
| Max Tree Depth           | 6                                  | 6                                         |
| Num. Leaves              | 31                                 | 31                                        |
| Subsample Ratio          | 0.8                                | 0.8                                       |
| Colsample By Tree        | 0.8                                | 0.8                                       |
| Minimum Child Weight     | 1                                  | 1                                         |
| Early Stopping Rounds    | 15                                 | 15                                        |

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**Table 8** List of Abbreviations

| Abbreviation | Definition                                                        | Abbreviation | Definition                    |
|--------------|-------------------------------------------------------------------|--------------|-------------------------------|
| CBD          | Central Business District                                         | MSE          | Mean Squared Error            |
| CPC          | Common Part of Commuters                                          | OD           | Origin-Destination            |
| GBDT         | Gradient Boosted Decision Tree                                    | OSM          | OpenStreetMap                 |
| GDPR         | General Data Protection Regulation                                | POI          | Point of Interest             |
| MLE          | Maximum Likelihood Estimation                                     | SHAP         | SHapley Additive exPlanations |
| SDF-ZTF      | Separable Decomposed Formulation for Zero-Shot Transferable Flows |              |                               |

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